Q1. ORM (Object-Relational Mapping) is a technique that maps database tables to Java objects, allowing developers to work with data using objects instead of SQL queries. **Advantages over JDBC:** Automatic table-to-object mapping, built-in caching, lazy loading, automatic connection management, and database independence (no vendor-specific SQL). *Example: In JDBC, you manually write `SELECT * FROM users` and parse ResultSet; in Hibernate, you just call `session.get(User.class, id)`.*

Q2. Hibernate is an open-source ORM framework for Java that simplifies database operations by mapping Java classes to database tables and providing APIs for CRUD operations. Its role is to act as a bridge between Java objects and relational databases, handling connection pooling, caching, and SQL generation automatically so developers focus on business logic.

Q3. JPA (Java Persistence API) is a specification (not implementation) that defines how to manage relational data in Java applications using annotations. **Key annotations in model class:**

- `@Entity` – Marks class as a database table
- `@Table(name="table_name")` – Specifies the actual table name
- `@Id` – Marks the primary key field
- `@Column(name="col_name")` – Maps field to a specific column
- `@GeneratedValue(strategy=GenerationType.IDENTITY)` – Auto-generates primary key values (e.g., auto-increment)